

B.Tech Degree IV Semester Examination, April 2010**ME 403 APPLIED THERMODYNAMICS**
(2002 Scheme)

Time: 3 Hours

Maximum Marks: 100

(All questions carry **EQUAL** marks)

- I. (a) State the two statements of second law of thermodynamics. Compare both, how they are similar.
- (b) An engine working on carnot cycle receives 1200 kJ from a heat source at a constant temperature of 1000°C and rejects to a heat sink at a constant temperature of 35°C. Calculate:
- (i) Thermal efficiency of the engine
- (ii) Work done

OR

- II. (a) Write notes on:
- (i) Thermodynamic temperature scale
- (ii) Entropy
- (iii) Enthalpy
- (b) During a process, temperature of a system rises from 100°C to 200°C, heat transfer per degree rise in temperature, at each temperature reached during the process is given by:

$$\frac{dQ}{dT} = 1.005 \text{ kJ/K}$$

The work done on the system per degree rise in temperature at each temperature reached is given by:

$$\frac{dW}{dT} = (4 - 0.2T) \text{ kJ/K}$$

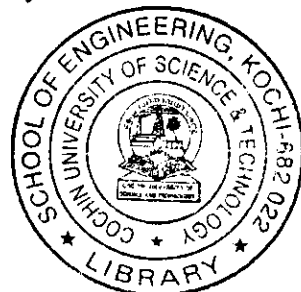
Calculate the change in internal energy of the system during the process.

- III. (a) State Dalton's law of partial pressure and Amagat's law of partial volume.
- (b) A mixture contains 2kg of He and 5kg of N₂ at 300k, 100KPa is compressed in a reversible adiabatic process to 800 KPa. Calculate:
- (i) The partial pressure of constituents at the end of compression
- (ii) Find the temperature (iii) Change in internal energy of the mixture.

		Cp	Cv
Sl.No.	Gas	KJ/kgK	KJ/kgK
1	N ₂	1.04	0.743
2	He	5.23	3.14

OR

- IV. A mixture of ideal gases consist of 2kg of N₂ and 4 kg of CO₂ at a pressure of 300 KPa, 25°C. Calculate:
- (i) The mole fraction of each constituent
- (ii) Molecular weight of mixture
- (iii) Gas constant of mixture
- (iv) Partial pressures and volumes
- (v) Volume and density of mixture
- (vi) Cp and Cv of mixture
- Take γ for CO₂ and N₂ to be 1.286 and 1.4 respectively.



(Turn Over)

- V. (a) Explain working of Bomb Calorie meter.
 (b) The flue gas analysis by volume for burning of C_6H_{14} is given below:
 $CO_2 = 8.5\%$, $CO = 7.8\%$ and $N_2 = 83.7\%$. Calculate:
 (i) The air fuel ratio required for chemically correct combustion
 (ii) A : F ratio as a percentage of chemically correct mixture A : F ratio.
- OR**
- VI. (a) Explain Orsat apparatus construction.
 (b) A coal has the following composition by mass $C = 0.89$, $H_2 = 0.03$, $S = 0.01$, $O_2 = 0.02$ and $N_2 = 0.03$ and remaining is ash. Calculate the HCV and LCV of coal.
- VII. (a) Draw value timing diagram and otto cycle of a 4 stroke engine.
 (b) A petrol engine with compression ratio of 5 develops 20 kW indicated power and consumes 8 litres of fuel per hour. The specific gravity of fuel is 0.78 and its calorific value is 44 MJ/kg. Calculate the indicated thermal efficiency and relative efficiency. Take $\gamma = 1.4$.
- OR**
- VIII. (a) Describe Morse Test.
 (b) In an engine, working on diesel cycle, the air fuel ratio by weight is 50 : 1. The temperature of air at the beginning of combustion is $40^\circ C$ and the compression ratio is 19. What is the ideal efficiency of the engine. Calorific value of fuel is 42 MJ/kg, assume $C_v = 0.717$ kJ/kgK, $C_p = 1.004$ kJ/kgK.
- IX. (a) Explain working of simple carburetor with a neat sketch.
 (b) What are the causes of detonation in an SI engine?
 (c) With the help of a diagram explain water cooling system in an IC engine.
- OR**
- X. Write notes on:
 (i) Combustion chambers of CI engine and SI engine.
 (ii) Turbo charging
 (iii) Wankel engine
 (iv) Additives in petrol